



University of
Nottingham
UK | CHINA | MALAYSIA

School of Mathematical Sciences

Autumn Semester 2025-2026

MATH4065 - STATISTICAL FOUNDATIONS

Assessed Coursework

Deadline: 3pm, Wednesday 3/12/2025

Your neat, clearly-legible solutions should be submitted electronically as a pdf file via the MATH4065 Moodle page by the deadline indicated there. A scan of a handwritten solution is acceptable. As this work is assessed, your submission must be entirely your own work (see [the University's policy on Academic Misconduct](#)); in particular, the University considers the unauthorised use of AI tools in assessments to constitute false authorship.

Submissions up to five working days late will be subject to a penalty of 5% of the maximum mark per working day.

Deadline extensions due to Support Plans and Extenuating Circumstances can be requested according to School and University policies, as applicable to this module.

Your marked coursework, the solutions (where appropriate) and the associated feedback will normally be released to you between 10 and 20 working days from the main cohort submission deadline for Level 0 and 1 modules, and between 10 and 15 working days from the main cohort submission deadline for higher level modules.

IMPORTANT: Your answers should include, where appropriate

- (i) any R code you used;
- (ii) any plots you produce;
- (iii) explanations of what you are doing and why.

1. For this question you will need the file "ExperimentData.csv" which can be downloaded from the module Moodle page. This file contains the results of an experiment in which the input variables (i.e. the predictor variables) are Type, Group, Input1, Input2 and Input3, and the output variable (i.e. the response variable) is Output.

(a) Read the data file into R and use appropriate R commands to see if there is any graphical evidence for relationships between Output and the five other variables. [5 marks]

(b) Using R, fit a linear model (model 1) in which Output is the response variable and the five other variables are all included as predictor variables as well as a constant term. Find the least-squares estimates of the model parameters. Find two of the five other variables which could sensibly be excluded, and fit a new linear model (model 2) with these variables removed. [10 marks]

(c) Explore how well model 2 fits the data by considering the residuals and looking for evidence that the residuals are normally-distributed. [5 marks]

2. Consider a random sample $X_1, \dots, X_n$ from the continuous distribution with probability density function

$$f(x) = \theta x^{\theta-1} \exp(-x^\theta), \quad x > 0,$$

where $\theta > 0$.

(a) Derive the log-likelihood function $l(\theta)$. [2 marks]

(b) Use R to find the maximum likelihood estimate of θ given observed data

$$X_1 = 0.610, X_2 = 0.309, X_3 = 0.578, X_4 = 0.921, X_5 = 1.04, \\ X_6 = 0.820, X_7 = 1.71, X_8 = 0.834, X_9 = 0.852, X_{10} = 1.11$$

[5 marks]

(c) Produce a plot of $l(\theta)$ over a suitable range of values with the maximum likelihood estimate of θ shown. [3 marks]

3. Let X and Y be independent random variables such that $X \sim N(0, 1)$ and $Y \sim N(1, 2)$. Let the random variable Z be defined, conditional on X , by $Z \sim \text{Exp}(|x|)$ if $X = x$, where $\text{Exp}(\lambda)$ denotes an exponential random variable with mean λ^{-1} .

Use simulation to provide numerical answers to the following questions. You will NOT be awarded marks for any other method.

- (a) Find $E\left[\sqrt{\left|\frac{X}{Y}\right|}\right]$. [2 marks]
- (b) Find $P(\sin(XY) > 0.5)$. [3 marks]
- (c) Find $P(X < Z)$. [5 marks]